JURONG JUNIOR COLLEGE
JC2 Preliminary Examination 2018

CANDIDATE NAME						
CLASS		INDEX NUMBER				

BIOLOGY **8876/02**

Paper 2 Structured Questions and Free-response Questions **23 August 2018**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Section B	
Total	

This document consists of **20** printed pages.

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows an electron micrograph of mitochondria cross-sections.

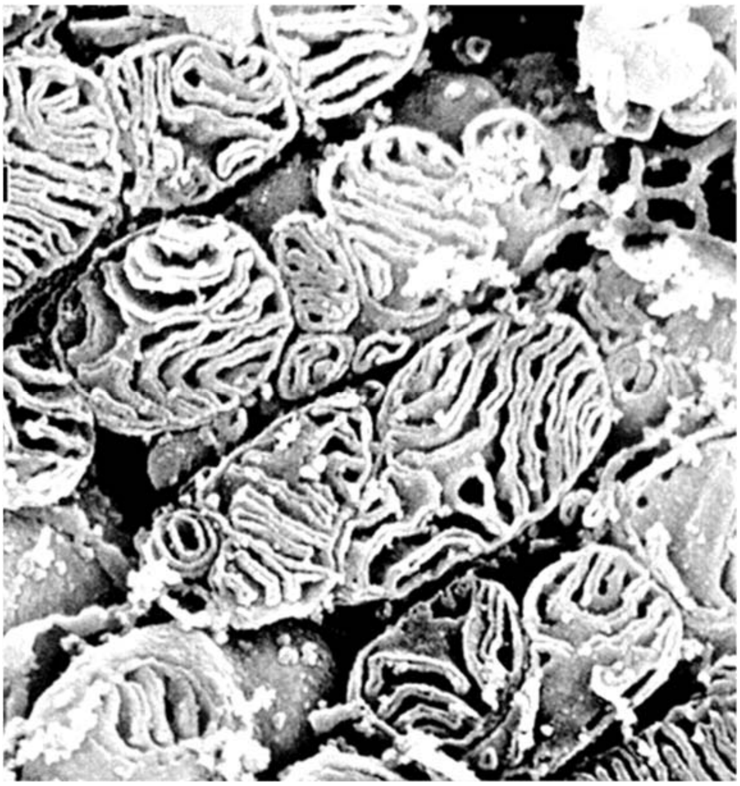

Fig. 1.1

(a) With reference to Fig. 1.1, state one visible feature of the mitochondrion and explain how this feature is adapted for the mitochondrion's function. [3]

Fig. 1.2 shows the molecular structure of carbonyl cyanide-4-(trifluoromethoxy) phenylhydrazone (FCCP). FCCP is a respiratory poison that binds protons and transports them across the phospholipid bilayer of the inner mitochondrial membrane. Protons alone are unable to diffuse freely in this manner.

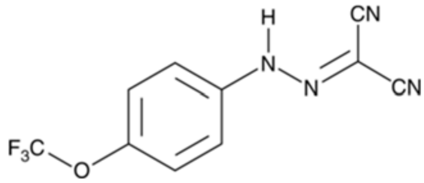

Fig. 1.2

(b) Explain, in relation to their properties, why FCCP readily diffuses across the phospholipid bilayer while protons do not. [2]

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In the presence of FCCP, the rate of ATP synthesis during respiration is significantly diminished. The rate is further reduced if oxygen becomes unavailable. In such conditions where oxygen is depleted, cells continue to oxidise glucose to form pyruvate and sustain ATP synthesis.

(c) (i) Identify the key stages where ATP synthesis occurs via substrate level phosphorylation. [1]

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(ii) State the location where the biochemical pathway enabling the continual oxidation of glucose to form pyruvate occurs. [1]

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(iii) Explain how ATP synthesis can be sustained in the absence of oxygen in yeast cells. [3]

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[Total: 10]

2 Fig. 2.1 shows a representation of a starch molecule. Starch is made up of amylose and amylopectin.

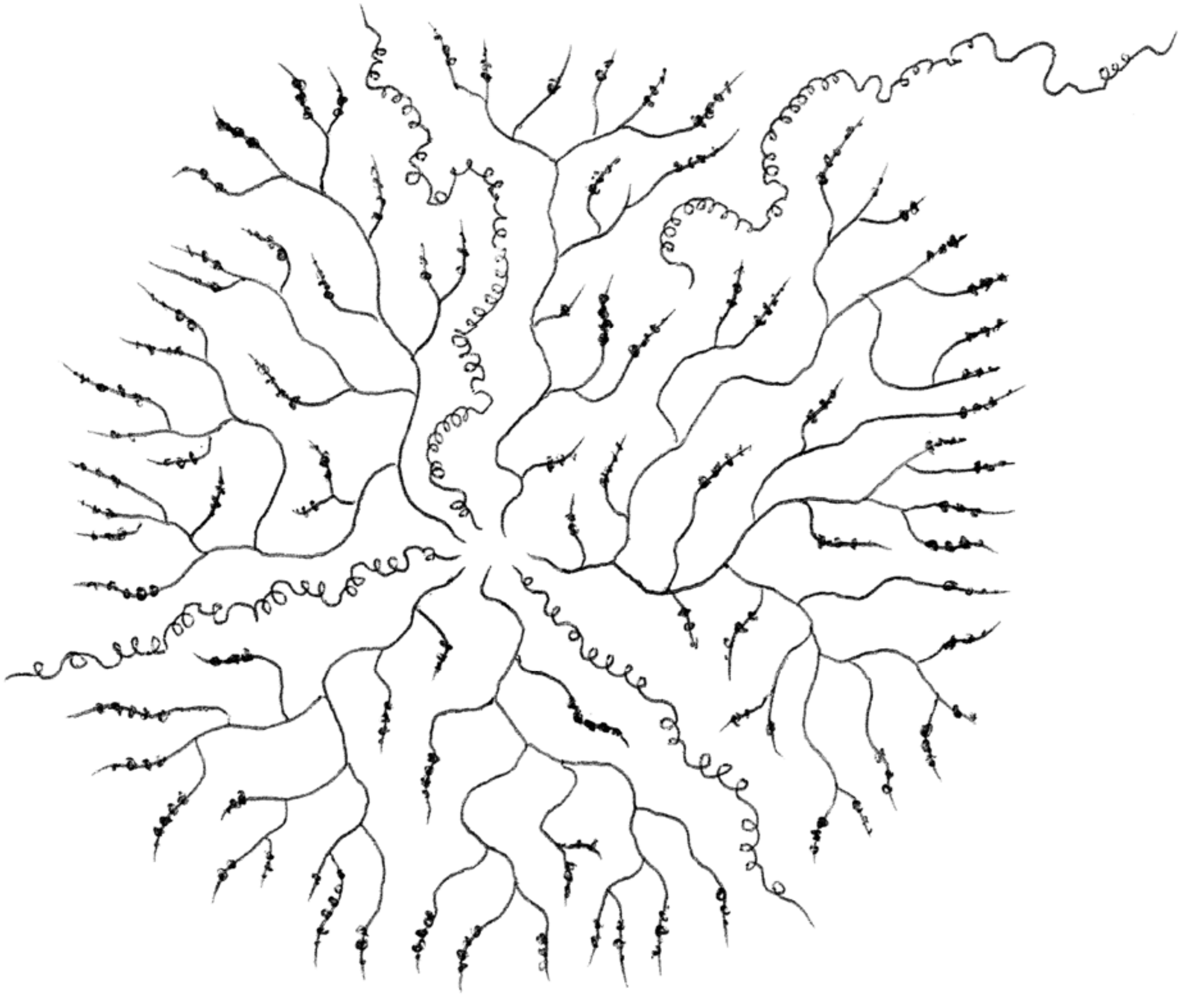

Fig. 2.1

(a) On Fig. 2.1, label amylose and amylopectin. [2]

(b) Explain how the structure of amylopectin is related to its role in living organisms. [2]

(c) Describe one structural difference between cellulose and amylopectin. [1]

[Total: 5]

3 Fig. 3.1 shows the origin and development of a B cell (a type of white blood cell) from a stem cell.

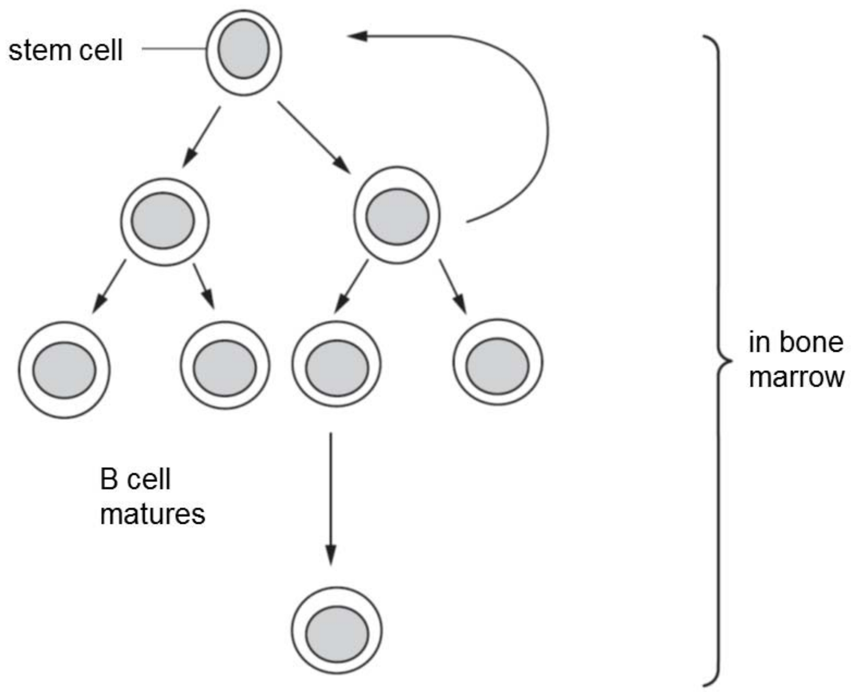

Fig. 3.1

(a) Explain what is meant by a *stem cell*. [2]

Turn over for the remainder of Question 3

During B cell development, specific cell surface proteins known as CD proteins are produced and incorporated into the cell surface membrane.

Fig. 3.2 shows the stages involved in the synthesis of a CD protein in a B cell.

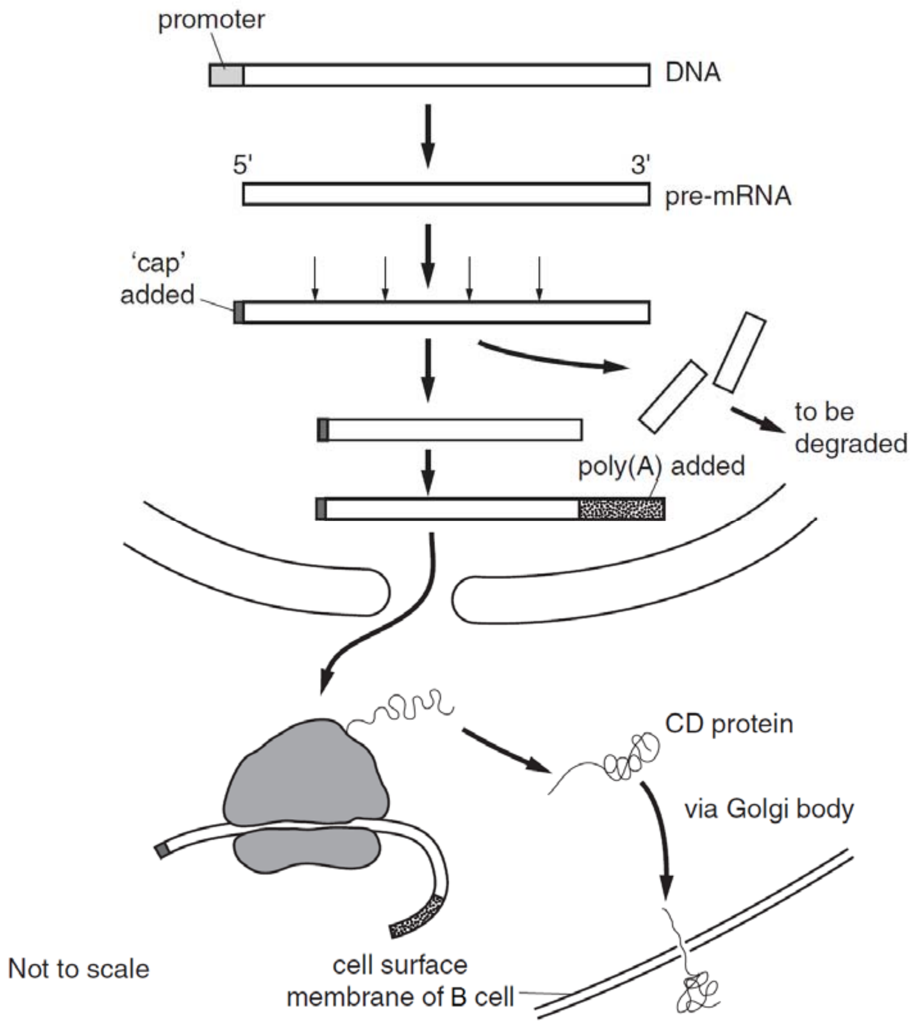

Fig. 3.2

(b) With reference to Fig. 3.2, outline the process:

(i) occurring in the nucleus of the B cell to produce pre-mRNA [3]

(ii) in forming the translation initiation complex. [3]

(c) Burkitt's lymphoma is a cancer of B cells. It is caused by a mutation that usually involves chromosome 8.

(i) Name one causative agent of mutation in chromosome 8 of Burkitt's lymphoma. [1]

(ii) Describe the key changes of two types of genes and their effects on cancer development. [2]

[Total: 11]

4 Galactosaemia is a rare genetic disease in which the build-up of the monosaccharide galactose can result in an enlarged liver, kidney failure and brain damage. Galactose is produced in the body from the digestion of the sugar lactose, found in milk.

Galactosaemia is caused by a recessive mutation of the *GALT* gene. The normal dominant allele codes for an enzyme which converts galactose to glucose.

(a) **(i)** Suggest how a person with galactosaemia can minimise damage to the liver, kidney and brain. [1]

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(ii) Explain how a mutation in the *GALT* gene could result in a change in the enzyme responsible for the metabolism of galactose. [4]

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(b) If the phenotypes of parents are known, the probabilities of having a child with galactosaemia, an unaffected child (healthy, not a carrier) or a child who is a carrier can be calculated.

Complete Table 4.1 to show the results of these calculations. [2]

Table 4.1

parent 1	**parent 2**	**percentage probability of having a child with galactosaemia**	**percentage probability of having an unaffected child**	**percentage probability of having a child who is a carrier**
unaffected	carrier	0	50	50
carrier	carrier	....	....	....
unaffected	has galactosaemia	....	....	....
carrier	has galactosaemia	50	0	50

[Total: 7]

5 Silvereyes (*Zosterops lateralis*) are a species of Australian songbirds that can be found in both rural and urban environments. These birds produce contact calls to inform one another of danger or newly discovered food sources. The pitches and speed (measured in number of syllables per second) of Silvereyes' contact calls were identified to be genetically associated.

Studies on Silvereyes living in increasingly denser Australian cities have revealed that the birds made contact calls with much higher mean pitches and slower mean speeds with longer pauses between calls over the years.

When measured in the past, the mean pitch of the Silvereye call overlapped significantly with the pitches of road and air traffic making the calls indistinguishable from urban noise. Sound-reflecting buildings also diminished the clarity of the bird calls.

Fig. 5.1 shows a Silvereye.

Fig. 5.1

(a) Explain how natural selection may have led to a change in the mean pitch and speed of the Silvereye calls in the city. [5]

(b) An analysis of ice cores from the Arctic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 5.1 shows the concentrations of carbon dioxide measured in ice cores, dated between 1000 and 2000 AD.

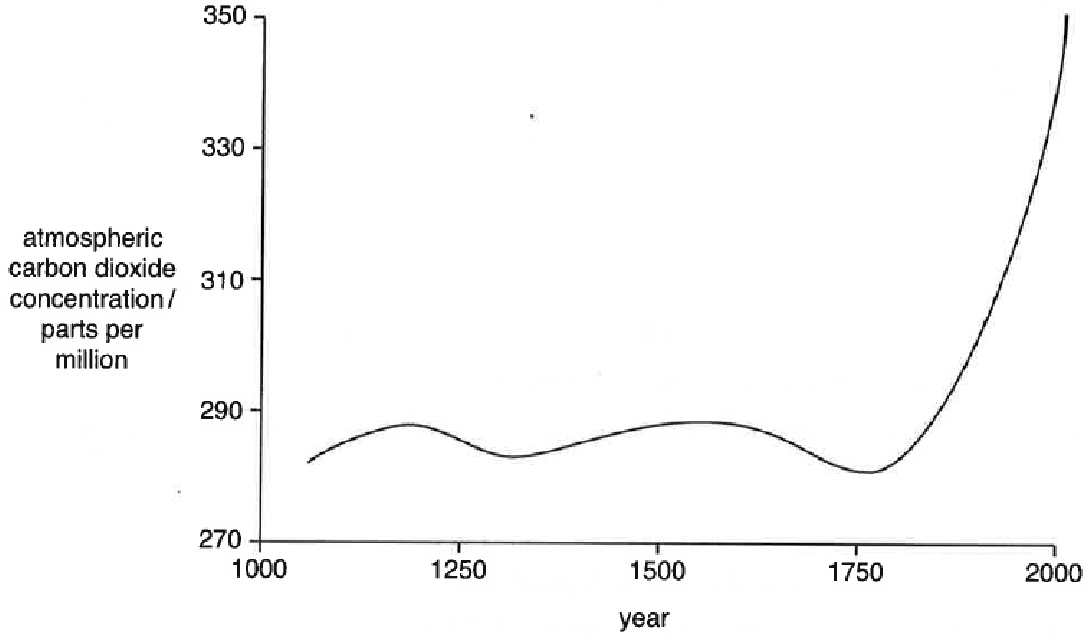

Fig. 5.1

(i) Describe the trend in Fig. 5.1. [2]

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(ii) Atmospheric carbon dioxide concentrations show regular annual variations. Suggest one reason for this. [1]

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(c) Fig. 5.2 shows that, over the same period of time, the average surface temperature of the Earth has shown a similar pattern of change. The increasing concentrations of carbon dioxide is thought to be responsible for the increase in temperature over the last 100 years. This is referred to as the enhanced greenhouse effect.

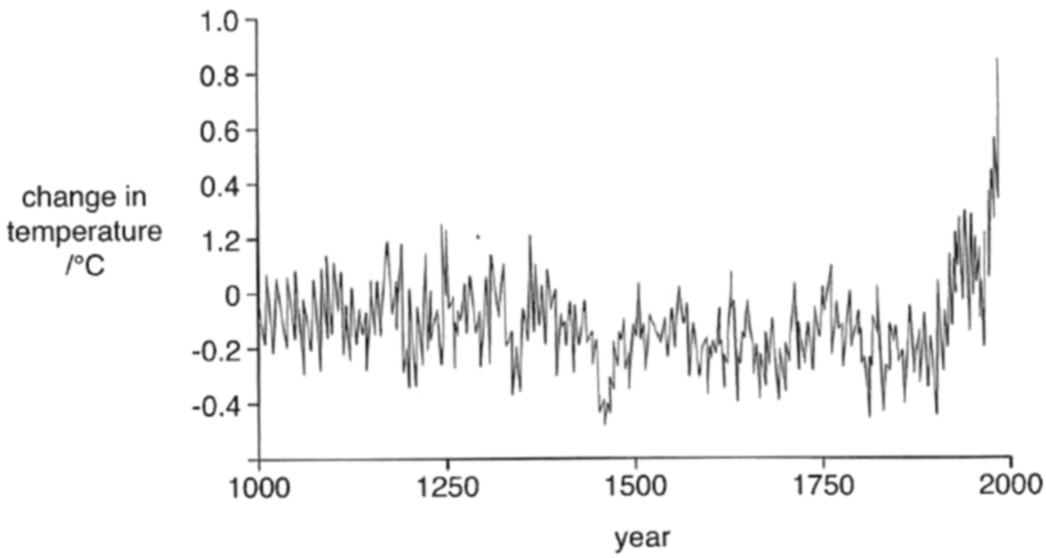

Fig. 5.2

(i) Describe one way in which the data in Fig. 5.2 resembles the data in Fig. 5.1 and one way in which it is different. [2]

Similarity ____

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Difference ____

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(ii) Explain the human activities that have contributed to the enhanced greenhouse effect. [2]

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[Total: 12]

Section B starts on page 16

Section B

Answer **one** question in this section

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

6 **(a)** The organelles of the endomembrane system in animal cells are related through direct contact or by the transfer of membrane segments as vesicles.

Outline the functions of the organelles of the endomembrane system and state the structural similarities between these organelles. [9]

(b) A 75 kg human consists of ten times more prokaryotic cells than eukaryotic cells, with a total prokaryote mass of at least 1 kg. This assembly of prokaryotic cells is known as the human prokaryotic microbiome community.

Outline the differences between typical prokaryotic and eukaryotic cells and state the methods by which these differences can be shown. [6]

[Total: 15]

7 **(a)** Variation exists in individuals of the same species in a population due to a number of different reasons. Describe what causes variation and why it is important in natural selection. [9]

(b) Discuss, using known examples, how limiting factors can influence the rate of various biological processes. [6]

[Total: 15]